DecodeLabs Internships 2026

DEVOPS PROJECT 2

Version Control with Git

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Submitted To

DecodeLabs

Project Details

Project Number: Project 2

Project Title: Version Control with Git

Technology Used: Git,Github,WSL(Ubuntu)

Submission Date: 11-06-2026

Declaration

I hereby declare that this project report titled **"Version Control with Git"** has been completed by me as part of the DecodeLabs Industrial Training Program. The work presented in this report is based on my own practical implementation and learning experience using Git and GitHub.

Project 2: Version Control with Git – Introduction

Version control is a fundamental practice in modern software development that enables developers to track changes, manage project history, and collaborate effectively on codebases. As software projects grow in size and complexity, maintaining different versions of files and coordinating changes among multiple developers becomes increasingly challenging. Version control systems solve this problem by providing mechanisms to record modifications, restore previous versions, and manage contributions from different team members. Among the various version control systems available today, Git has emerged as the most widely used and industry-standard solution.

Git is a distributed version control system developed to provide speed, reliability, and flexibility in managing software projects. Unlike traditional centralized systems, Git allows every developer to maintain a complete copy of the repository, including its entire history. This distributed architecture enables developers to work independently, make changes locally, and synchronize their work with remote repositories whenever necessary. Git provides powerful features such as branching, merging, staging, and commit tracking, making it an essential tool for modern software development and DevOps workflows.

GitHub is a cloud-based platform built around Git that enables developers to host repositories, collaborate on projects, review code, and manage software development activities. GitHub serves as a central location where developers can store source code, track issues, manage pull requests, and automate development processes. It has become one of the most popular platforms for open-source development and professional software engineering. In DevOps environments, GitHub plays a critical role in supporting Continuous Integration (CI), Continuous Deployment (CD), and collaborative development practices.

One of the key advantages of Git is its ability to maintain a complete history of changes made to a project. Every modification can be tracked through commits, which act as snapshots of the project at specific points in time. Developers can review previous versions, identify changes, and revert to stable states when necessary. This capability significantly improves project reliability and reduces the risk of losing important work. Furthermore, Git facilitates teamwork by allowing multiple developers to work on different features simultaneously and merge their contributions efficiently.

This project focuses on understanding the fundamental concepts of Git and GitHub through practical implementation. The project involves creating a local Git repository, initializing version control, tracking file changes, staging files, creating commits, viewing repository history, connecting to a remote GitHub repository, and pushing code to the cloud. These activities demonstrate the complete workflow used by developers to manage source code effectively and maintain project integrity.

Through this project, valuable hands-on experience is gained in source code management and collaborative development practices. The knowledge acquired serves as a foundation for more advanced DevOps concepts such as automated pipelines, continuous integration, deployment automation, and infrastructure management. Mastering Git and GitHub is an essential step toward becoming proficient in modern software development and DevOps engineering.

Objective

The objective of this project is to understand and implement the fundamentals of Version Control using Git and GitHub. The project focuses on creating a local Git repository, tracking changes in files, creating commits, connecting the local repository to GitHub, and pushing code to a remote repository. Through this project, I aim to develop practical knowledge of source code management, collaboration workflows, and version tracking, which are essential skills in DevOps and modern software development.

Step 1: Verify Git Installation

```
jania_jose_kannath@JANIAJOSE:~$ git --version
git version 2.43.0
jania_jose_kannath@JANIAJOSE:~$
```

Step 2: Create Project Directory

```
jania_jose_kannath@JANIAJOSE:~$ mkdir Git_Project2
jania_jose_kannath@JANIAJOSE:~$ cd Git_Project2
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ pwd
/home/jania_jose_kannath/Git_Project2
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 3: Initialize Git Repository

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/jania_jose_kannath/Git_Project2/.git/
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 4: Check Repository Status

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 5: Create Project File

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ echo "# Git Project" > README.md
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ ls
README.md
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 6: Check Status Again

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md

nothing added to commit but untracked files present (use "git add" to track)
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 7: Add File to Staging Area

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git add README.md
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md

jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 8: Create First Commit

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git commit -m "Initial commit"
[master (root-commit) 8c1ea87] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 9: View Commit History

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git log --oneline
8c1ea87 (HEAD -> master) Initial commit
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

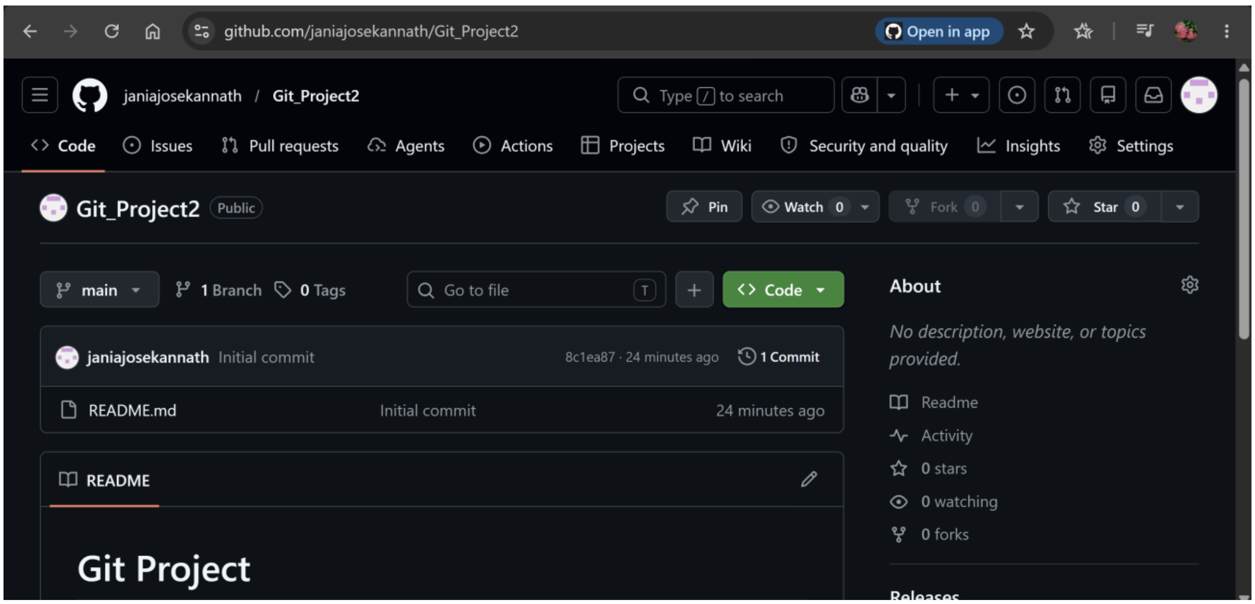
Step 10: Connect to GitHub Repository

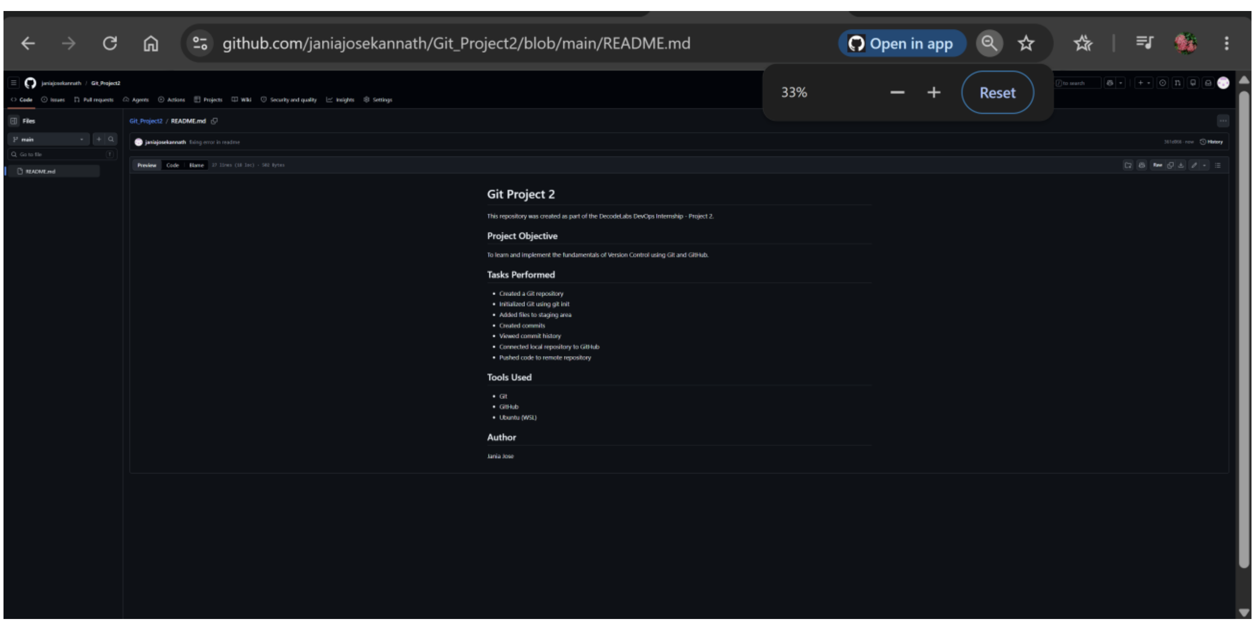
```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git remote add origin https://github.com/janiajosekannath/Git_Proje
ct2.git
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git remote -v
origin  https://github.com/janiajosekannath/Git_Project2.git (fetch)
origin  https://github.com/janiajosekannath/Git_Project2.git (push)
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 11: Push Code to GitHub

```
jania_jose_kannath@JANIAJOSE:~/Git_Project2$ git push -u origin main
Username for 'https://github.com': janiajosekannath
Password for 'https://janiajosekannath@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 231 bytes | 115.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/janiajosekannath/Git_Project2.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
jania_jose_kannath@JANIAJOSE:~/Git_Project2$
```

Step 12: Verify Repository on GitHub





Conclusion

Github Repository Link-https://github.com/janiajosekannath/Git_Project2

In this project, I successfully learned and implemented the basic Git workflow using Git and GitHub. I created a local repository, tracked file changes, staged files, created commits, viewed commit history, and connected the repository to GitHub for remote version control. This hands-on experience improved my understanding of source code management, version tracking, and collaboration practices used in DevOps and software development environments. The project provided a strong foundation for managing code efficiently and working with distributed development teams.